MRUNAL TALIKOTI

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PROFESSIONAL SUMMARY

B.Tech Computer Engineering student (CGPA 9.22/10, graduating 2027) and AI & Cloud-focused software developer with four internships building production AI pipelines on Azure, Docker and FastAPI. Built LLM document-processing systems handling 100,000+ documents per day at 89% extraction accuracy, and automated workflows that cut manual effort by 70–90%. Indian patent holder (multi-agent system generation), presented researcher, and hackathon winner (1st among 150+ teams). Skilled in Python, LangChain/LangGraph, RAG, vector databases, and clean, scalable code.

SKILLS

AI / LLM: Generative AI, RAG, Agentic AI, LLM, NLP, Prompt Engineering, LLM Evaluation, Fine-Tuning (QLoRA), Machine Learning, Deep Learning
Languages & Frameworks: Python, C++, PyTorch, LangChain, LangGraph, MCP, OpenAI SDK, Azure OpenAI, Hugging Face Transformers, Unsloth, FastAPI, Smolagents, Stable Diffusion, Pydantic
Vector Databases: FAISS, Pinecone
Cloud & DevOps: Azure (Document Intelligence, OpenAI), Docker, CI/CD, Celery, RabbitMQ, Redis, AWS S3, System Design, Scalable Architecture
Data & Databases: ETL Pipelines, Data Extraction, REST APIs, Automation, SQL, MySQL, PostgreSQL, MongoDB
Tools: GitHub (Code Reviews, Version Control), Jupyter, Google Colab, Supabase, n8n

EDUCATION

Vishwakarma Institute of Information Technology — B.Tech, Computer Engineering | CGPA: 9.22/10 *Expected 2027*

EXPERIENCE

Software Engineer Intern (Full Time) | BizAmica *Pune, Maharashtra | Jan 2026 – Aug 2026*

- Engineered document classification, verification, auto-allocation and extraction pipelines using LangChain and Pydantic, processing 100,000+ documents per day across 25+ document types.
- Integrated Azure Document Intelligence and Azure OpenAI LLMs to extract data from unstructured business documents at 89% accuracy and ~9 seconds per page, cutting manual processing time by 82%.

Software Intern (Part Time) – Industry Project | UST Global *Pune, Maharashtra | Jan 2026 – Apr 2026*

- Engineered an AI-driven analog workflow automation pipeline for transistor parameter prediction, SPICE netlist generation and circuit verification, cutting manual design time by 70%.
- Trained Random Forest Regressor (91.2% accuracy) and XGBoost models across 20+ transistor parameters; applied PSO for optimal results.

AI Intern | IoTIoT *Pune, Maharashtra | Jul 2025 – Dec 2025*

- Built a proof-of-concept multi-agent CRM automation using LangGraph and an MCP server, automating 6+ workflow steps.
- Designed autonomous SOP prototypes with LangChain and Smolagents, taking them from concept to working demo.

Machine Learning Intern | Valentius Kryptix *Pune, Maharashtra | Jul 2024 – Mar 2025*

- Built a Privacy.AI web UI (HTML/CSS/JS, Lightning Web Components) and automated legacy-system data extraction with SQL, cutting manual data entry by 90%.

PROJECTS

Meta-Agent (Indian Patent) *AnthropicModels, JWT+OAuth, Celery, Docker, DAG, Alembic, Redis, S3*

- Designed hierarchical orchestration of multiple sub-agents to generate agents from user prompts.
- Built a context manager for load balancing and a DAG-based execution manager, with object storage for artifacts and code-generator packages.

Clover — Fine-Tuning Studio | Winner, CodeApex 2026 *Python, FastAPI, React, Supabase, QLoRA, Unsloth, Ollama*

- Won 1st place among 150+ teams as part of a 4-member team with a no-code AI studio that lets domain experts fine-tune small language models (Qwen2.5-3B) using QLoRA and Unsloth (training loss 0.97, perplexity 2.64).
- Architected a FastAPI + React + Supabase platform with Ollama-based local inference for end-to-end model training, evaluation and deployment without ML expertise.

Factless *Sentence Transformers, FastAPI, Gemini, LangChain, Python SDK*

- Built a post-generation hallucination risk assessment system using a 7-layer agent architecture.
- Implemented semantic-search-based, color-coded risk scoring for hallucination detection in LLM outputs.

PUBLICATIONS, PATENT & ACHIEVEMENTS

- Indian Patent No. 202621005880 — "A System and a Method for Generation of an Agent"
- IEEE ICDBS — "Factless: Explainable Hallucination Detection" (Paper ID 1324) — presented, publication pending
- ICINDIA — "Vision Based Allergen Detection" (Paper ID 749) — presented, publication pending
- Partnered with FORTEA.IA on a machine learning approach for a Hemiplegia detection model.
- Certifications: Generative AI (Microsoft), AI Agents (Hugging Face), RAG Systems